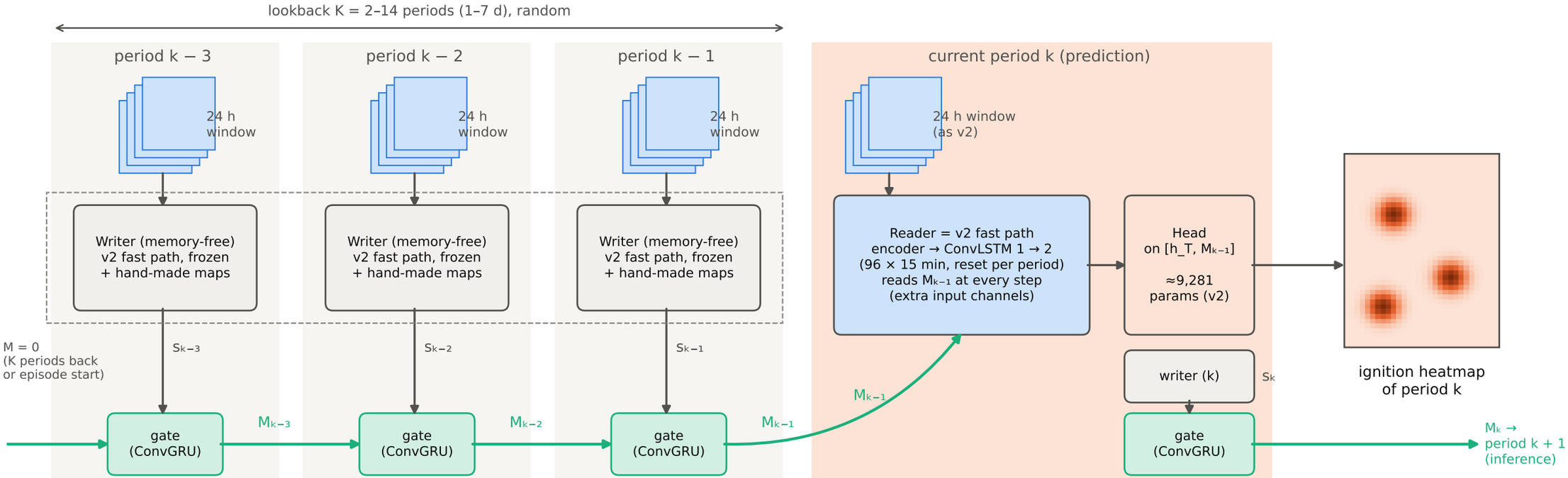
Concept: two timescales with a gated long-term memory (notes.md 6.6, Option 4; not implemented)

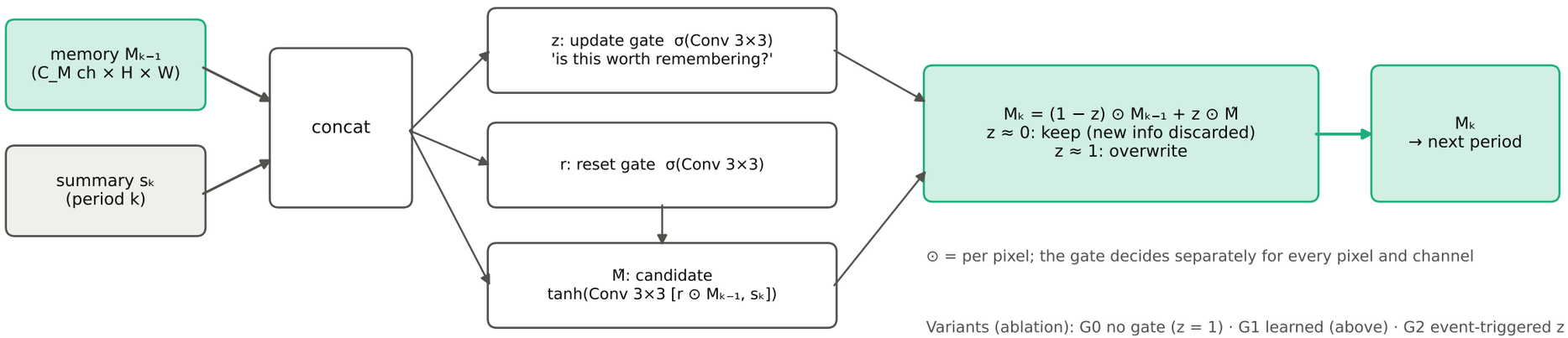
(a) Unrolled over consecutive 12 h periods: one training sample = current period k + lookback of K periods



gradients: reader, head and the K cheap gate steps (writer frozen; memory branch initialised to zero → starts exactly at v2)

writer summaries s: computed once on the full grid and cached (no gradient), then cropped and flipped with the sample

(b) The long-term gate: one ConvGRU step per period (Ballas et al., 2016)



⊙ = per pixel; the gate decides separately for every pixel and channel

Variants (ablation): G0 no gate (z = 1) · G1 learned (above) · G2 event-triggered z = aₖ ⊙ z (aₖ = detections nearby), else keep / decay γ^Δt · G3 sparse (L1 on z) · G4 several fixed rates. Dilated / stride-2 convs widen the memory's reach.

Gate maps z are an explainability output: where and when the model wrote to its long-term memory.